Fiscal Year:	FY 2011	**Task Last Updated:**	FY 05/04/2011
PI Name:	Kucik, Dennis F. M.D., Ph.D.		
Project Title:	Effect of high energy particle irradiation on adhesiveness of vascular endothelium and its consequences for atherosclerosis		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kucik@uab.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	205-934-0062
Organization Name:	University of Alabama at Birmingham		
PI Address 1:	Pathology		
PI Address 2:	1025 18th St S, CBSE room 239		
PI Web Page:			
City:	Birmingham	**State:**	AL
Zip Code:	35205-4831	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	02/01/2007	**End Date:**	01/31/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: changed to Cardiovascular Alterations Team (from Radiation Effects Team) effective 5/1/2008, per K. Major/NSBRI (6/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kabarowski, Janusz (University of Alabama at Birmingham)		
Grant/Contract No.:	NCC 9-58-RE00001		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Inflammation plays a major role in radiation damage to tissues. Chronic, low-level vascular inflammation is a risk factor for atherosclerosis. Consistent with this, exposure to radiation from a variety of sources is associated with increased risk of heart disease and stroke. Part of the inflammatory response to radiation is a change in the adhesiveness of the endothelial cells that line the blood vessels, triggering inappropriate accumulation of white blood cells, leading to later, damaging effects of inflammation. Although some studies have been done on the effects of gamma irradiation on vascular endothelium, the response of endothelium to the high energy radiation likely to be encountered in prolonged space flight had not previously been determined. Our hypothesis was that high energy heavy ions altered the adhesive properties of vascular endothelium, and resultant vascular inflammation accelerated atherosclerosis. We tested this hypothesis using both isolated cells and whole mice to determine the effects on vascular endothelium and the consequences for development of atherosclerosis. The original objectives were: Aim #1. Determine the effects of iron ion and proton irradiation vs. gamma irradiation on expression of key adhesion molecules on cultured endothelial cells, both at baseline and following cytokine stimulation. Aim #2. Determine the consequences (for adhesion) of iron ion- and proton-irradiation-induced adhesion molecule expression and compare these to gamma irradiation using functional adhesion assays. Aim #3. Determine whether radiation accelerates atherosclerosis in vivo using the well characterized ApoE mouse model. Aim #4: Determine whether Tepoxalin reduces radiation-induced atherosclerosis. Key Findings: 1) X-rays (substituted for gamma rays, which are very similar), protons and iron ions did not increase surface expression or endothelial adhesion molecules as much as typical inflammatory stimuli, such as TNF-alpha. Since endothelial cell adhesiveness is strongly increased in response to all types of radiation tested (see #2), mechanisms other than increased adhesion molecule expression are responsible for the adhesion change. Impact: This constitutes completion of Aim #1. Expression of key adhesion molecules is only weakly induced by radiation, in contrast to the effect of typical inflammatory stimuli. This has led us to investigate other mechanisms for radiation-induced cell adhesion increases. 2) Adhesiveness of irradiated endothelial cells following X-ray, iron ion (56Fe) or proton irradiation was measured using a flow-chamber-based adhesion assay designed to mimic conditions in the bloodstream. In all cases, radiation increased adhesion. These adhesion increases were as pronounced as those produced by typical inflammatory stimuli, even though radiation-induced increases in adhesion molecule expression were typically much smaller. Impact: This confirms the hypothesis that radiation alters adhesive properties of vascular endothelium, and represents completion of Aim #2. This has significant implications for astronaut health on prolonged deep-space missions, since increased vessel wall adhesiveness is an early step in development of atherosclerosis. 3) To better understand the mechanism of the increased adhesion, we identified the endothelial adhesion molecules responsible: ICAM-1 and VCAM-1. These adhesion molecules are known to be important in the pathogenesis of atherosclerosis. Antibody blocking of integrin receptors for these adhesion molecules on leukocytes modulated the pro-adhesive effect of radiation. Impact: This provides an opportunity to identify possible therapeutic targets that may be amenable to countermeasures. 4) Animal experiments confirmed that radiation-induced changes have consequences for atherosclerosis. ApoE -/- mice were irradiated using collimated beams focused on the aortic arches and carotids. Iron-ion irradiation (600 MeV at 2 and 5 Gy) was found to accelerate development of atherosclerotic plaques in portions of the aortas and carotids of irradiated mice at 13 weeks post-irradiation as compared to un-irradiated controls. This confirms the hypothesis that components of cosmic radiation can accelerate atherosclerosis, and completes the original objective of Aim #3 for iron. Results of proton and silicon ion experiments are being analyzed. Impact: 56Fe accelerates atherosclerosis and may pose a risk for astronaut health. Proposed Research for the coming year: This project is now complete. Research to determine mechanisms, early events, and dose dependence of radiation-induced atherosclerosis will continue as a new NASA project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Among the non-cancer risks of radiation exposure are inflammatory diseases in general and atherosclerosis in particular. Increased risk of atherosclerosis is a known side effect of radiation exposure on Earth, and increased endothelial adhesiveness (an inflammatory process) is an important component of the mechanism. In this project, we determined that 56Fe and proton radiation pose vascular disease risks and have begun to understand the mechanism. It is anticipated that a greater understanding of the risks posed by terrestrial radiation will also result. Insights gained in this project may also help to prevent, counter or ameliorate biological damage from terrestrial radiation sources such as nuclear weapons, nuclear power plant accidents, or radiation bioterrorism as well as adverse effects from therapeutic radiation. Additionally, knowledge gained from these experiments may lead to the use of in vitro assays to predict the risk of longer-term vascular damage.

Task Progress:	1) Flow-chamber based adhesion assay experiments were conducted using X-rays at UAB and 200 MeV protons and 600 MeV 56Fe at NSRL. This was to build on earlier experiments that had demonstrated increased adhesiveness of human aortic endothelial cells in response to radiation. Dose-response curves were refined and the character of the adhesive interactions (rolling vs. firm adhesion) was analyzed. It was found that X-rays, iron ions and protons all have the potential to increase vessel wall adhesiveness, which is an early step in development of atherosclerosis. 2) Mechanisms of radiation-induced adhesiveness changes were explored and it was determined that, unexpectedly, adhesion molecule expression changes play a minor role. However, cell signaling between endothelial cells and monocytic cells was occurring 24 hours after irradiation, and blocking particular pathways abrogated radiation-induced adhesion. Moreover, increased adhesiveness involved activation of specific integrins. Specific signaling molecules are being identified as we begin the next phase of this project. 3) A manuscript detailing results of X-ray experiments was submitted to Radiation Research. After review, the paper is now under revision. A manuscript describing 56 Fe results is in preparation. 4) Animal experiments confirmed that radiation-induced changes have consequences for atherosclerosis. ApoE -/- mice were irradiated using collimated beams focused on the aortic arches and carotids. Mice were given 600 MeV 56Fe at 2 and 5 Gy or 200 MeV protons at 0.5, 2, or 5Gy. Mice were then sacrificed at 13 and 40 weeks post-irradiation to determine whether iron ion and/or proton radiation accelerated development of atherosclerosis. We found that radiation significantly accelerates development of atherosclerosis in this mouse model. Results of the 56Fe experiments have been accepted for publication in Radiation Research. In addition, a further analysis of the character of the atherosclerotic lesions has been submitted for publication to Gravitational and Space Biology.
Bibliography Type:	Description: (Last Updated: 04/12/2018)
Articles in Peer-reviewed Journals	Khaled S, Gupta K, Wu X, Yu T, Chang P, Kabarowski JHS, Kucik DF. "Radiation increases endothelial cell adhesiveness through a chemokine-dependent mechanism." Submitted to Radiation Research, as of March 2011. , Mar-2011
Articles in Peer-reviewed Journals	Yu T, Parks BW, Yu S, Srivastava R, Gupta K, Wu X, Khaled S, Chang PY, Kabarowski JH, Kucik DF. "Iron-ion radiation accelerates atherosclerosis in apolipoprotein E-deficient mice." Radiat Res. 2011 Apr 5. [Epub ahead of print] PMID: 21466380 , Apr-2011
Articles in Peer-reviewed Journals	Yu T, Parks BW, Yu S, Srivastava R, Gupta K, Wu X, Khaled S, Chang PY, Kabarowski JH, Kucik DF. "Iron ion (56Fe) irradiation increases the size of pre-existing atherosclerotic lesions in apoE -/- mice." Submitted to Gravitational and Space Biology, as of 03/2011. , Mar-2011